Apportionment as Plane Geometry: Hamilton and Webster on the Simplex

Throughout this worksheet we will be working on the quota space (also called a simplex) $\mathcal{Q} = \{(x, y, z) \in \mathbb{R}^3_{\geq 0} : x + y + z = 4\}$. Our apportionment space is $\mathcal{M} = \{(x, y, z) \in \mathbb{Z}\mathbb{Z}^3_{\geq 0} : x + y + z = 4\}$ (the subset of points in \mathcal{Q} with integer coordinates).

Barycentric Coordinates and the Simplex. The plane slice $x+y+z = 4$ in the positive orthant is represented by an equilateral triangle. Each point in the triangle corresponds to a triple with nonnegative coordinates summing to 4. The vertices correspond to $(4, 0, 0)$, $(0, 4, 0)$, and $(0, 0, 4)$.

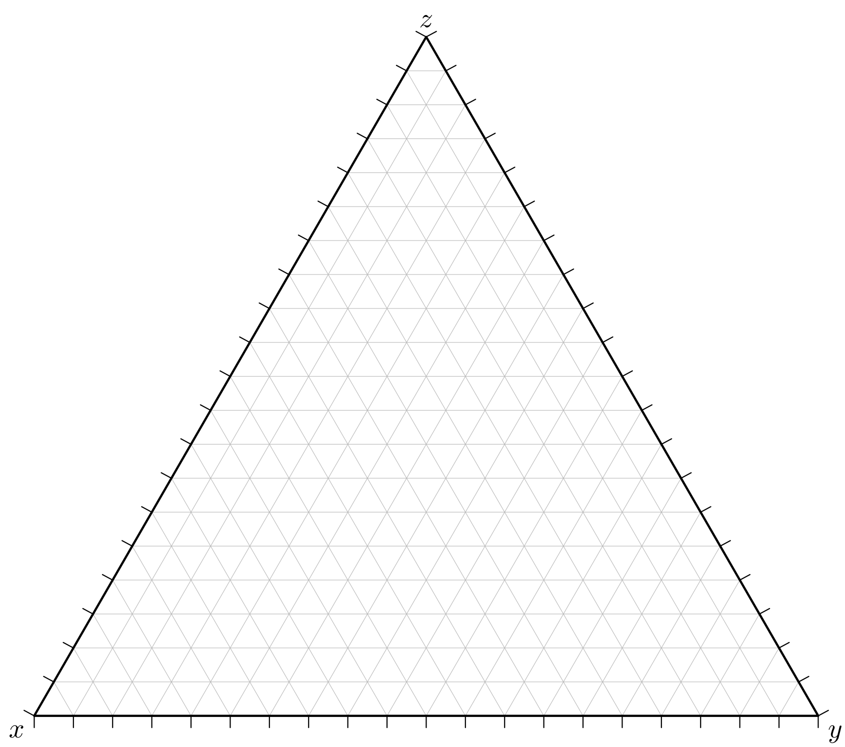

Figure 1 Barycentric grid for $x + y + z = 4$.

Checkpoint 2 Mark each point on the barycentric triangle and explain how you located it:

$(2, 1, 1)$, $(3, 1, 0)$, $(1, 2, 1)$.

Hint.

Checkpoint 3 How many integer triples satisfy $x + y + z = 4$?

How many of these lie on the boundary of the simplex? How many are strictly interior?

Hamilton's Method as a Geometric Function. Hamilton's method assigns seats by first taking lower quotas and then distributing the remaining seats according to the sizes of the fractional parts (remainders). In this section, you will build Hamilton regions step by step.

Recall, our quota space is $\mathcal{Q} = \{(x, y, z) \in \mathbb{R}^3_{\geq 0} : x + y + z = 4\}$.

Definition 4 Hamilton's Method.

1. Compute the lower quota vector $\ell = (\lfloor x \rfloor, \lfloor y \rfloor, \lfloor z \rfloor)$.
2. Let $s = 4 - (\ell_1 + \ell_2 + \ell_3)$ be the number of surplus seats.
3. Let $r_i = q_i - \lfloor q_i \rfloor$ be the remainders. Let $S(q)$ be the set of indices corresponding to the s largest remainders.
4. Define $H(q) = \ell + \sum_{i \in S(q)} e_i$.

$\diamondsuit$

Step 1: Floor Slabs. The first step of Hamilton's method depends only on the floor function. This divides the simplex into "slabs" where the lower quota vector ℓ is constant.

Checkpoint 5 Suppose $q = (x, y, z)$ satisfies

$$1 \leq x < 2, \quad 0 \leq y < 1, \quad 1 \leq z < 2.$$

(a) Compute the lower quota vector ℓ.
(b) Compute the number of surplus seats s.

Solution. (a) Since $1 \leq x < 2$, we have $\lfloor x \rfloor = 1$. Since $0 \leq y < 1$, we have $\lfloor y \rfloor = 0$. Since $1 \leq z < 2$, we have $\lfloor z \rfloor = 1$. Therefore $\ell = (1, 0, 1)$.

(b) The total number of seats is 4. The lower quotas sum to $\ell_1 + \ell_2 + \ell_3 = 1 + 0 + 1 = 2$, so the number of surplus seats is $s = 4 - 2 = 2$.

Checkpoint 6 Describe the shape of this region inside the simplex. Is it a triangle, a quadrilateral, or something else?

Solution. Inside the plane $x+y+z = 4$, the inequalities $1 \leq x < 2$, $0 \leq y < 1$, $1 \leq z < 2$ carve out a polygonal region (a slice of a rectangular box by a plane). In this particular case it is a region that looks like a *triangle* missing parts of its boundary.

One way to see this is to eliminate z using $z = 4-x-y$. Then the conditions become $1 \leq x < 2$, $0 \leq y < 1$, and $1 \leq 4 - x - y < 2$, i.e. $2 < x + y \leq 3$. So in the (x, y)-plane we are intersecting the rectangle $[1, 2) \times [0, 1)$ with the strip $2 < x + y \leq 3$, which produces a shape that is morally a triangle.

Now we can write the remainders using the lower quota vector $\ell = (1, 0, 1)$. By definition $r_i = q_i - \ell_i$, so $r_1 = x - 1$, $r_2 = y$, and $r_3 = z - 1$.

Checkpoint 7 Suppose Hamilton assigns surplus seats to coordinates 1 and 2. Write the inequalities that express this condition in terms of remainders.

Then rewrite these inequalities using x, y, z and the relation $x + y + z = 4$ (which lets you eliminate z).

Solution. Since there are $s = 2$ surplus seats in this slab, coordinates 1 and 2 get the surplus seats means that their remainders are at least as large as the remaining remainder:

$$r_1 \geq r_3, \qquad r_2 \geq r_3.$$

Using $r_1 = x - 1$, $r_2 = y$, and $r_3 = z - 1$, these become

$$x - 1 \geq z - 1 \quad \text{and} \quad y \geq z - 1,$$

i.e.

$$x \geq z \quad \text{and} \quad y + 1 \geq z.$$

Using $z = 4 - x - y$, the inequalities can be written purely in x, y:

$$x \geq 4 - x - y \iff 2x + y \geq 4,$$

$$y + 1 \geq 4 - x - y \iff x + 2y \geq 3.$$

(Together with the slab constraints, these describe the $S = \{1, 2\}$ piece.)

Step 3: One Hamilton Piece. We now combine the slab conditions and remainder inequalities to form a single Hamilton piece.

Checkpoint 8 Let $m = (2, 1, 1)$ and consider the surplus set $S = \{1, 2\}$.

1. Compute the corresponding lower quota vector ℓ.
2. Write the slab inequalities for this choice of ℓ.
3. Write the remainder comparison inequalities.

Checkpoint 9 On the barycentric grid:

- Draw the slab region.
- Draw the remainder-equality boundaries.
- Shade the region corresponding to this S-piece.

Step 4: Building the Full Hamilton Region. A single allocation m may arise from several different surplus sets S. The full Hamilton preimage is the union of all corresponding pieces.

Checkpoint 10 For $m = (2, 1, 1)$, list all possible surplus sets S.

Checkpoint 11 Sketch all pieces corresponding to these surplus sets on the same barycentric triangle.

What do you notice about how the pieces fit together?

Step 5: Interpreting Hamilton's Geometry.

Checkpoint 12 Explain why Hamilton regions are intersections of:

- floor slabs, and
- halfspaces coming from remainder comparisons.

Checkpoint 13 Describe what happens to the Hamilton output when a point crosses a remainder boundary. What changes? What stays the same?

Webster's Method and Divisor Geometry. Webster's method is a divisor method. Unlike Hamilton's method, which works directly with the quota vector q, Webster introduces a global scaling factor called the **divisor**.

We begin with a population vector $p = (p_1, p_2, p_3)$ and a seat total $M = 4$. For a chosen divisor $d > 0$, we define scaled quotas

$$q_i = \frac{p_i}{d}.$$

Webster then rounds each q_i to the nearest integer.

Definition 14 Webster Rounding Rule.

$$\mathrm{W}(u) = \lfloor u + \tfrac{1}{2} \rfloor.$$

(When u is exactly halfway between two integers, a tie-breaking convention is required. In this activity we will avoid ties.) ◊

Step 1: Rounding Bands for One Coordinate. First, focus on just one coordinate. Suppose Webster outputs $m_i = 2$.

Checkpoint 15 Write the inequality that characterizes when $\mathrm{W}(q_i) = 2$.

Express your answer as a double inequality of the form

$$\underline{\hspace{2cm}} \ \le \ q_i \ < \ \underline{\hspace{2cm}}.$$

Checkpoint 16 Interpret this inequality geometrically. What kind of region does it describe on the number line?

Step 2: Divisor Intervals for a Fixed Allocation. Now suppose Webster outputs the allocation $m = (2, 1, 1)$.

Checkpoint 17 For each coordinate, write the rounding condition in terms of d.

For example, for the first coordinate:

$$2 - \tfrac{1}{2} \ \le \ \frac{p_1}{d} \ < \ 2 + \tfrac{1}{2}.$$

Solve this inequality for d.

Repeat this process for the second and third coordinates.

Checkpoint 18 You should now have three intervals for d, one from each coordinate.

Explain why Webster can output $m = (2, 1, 1)$ if and only if these three intervals overlap.

Checkpoint 19 What does it mean geometrically if the three divisor intervals do not overlap?

Step 3: From Divisor Intervals to Geometry on the Simplex. Instead of working directly with populations p, we can think in terms of scaled quotas $q = p/d$.

For Webster to output m, we must have

$$m_i - \tfrac{1}{2} \ \le \ q_i \ < \ m_i + \tfrac{1}{2} \quad \text{for each coordinate.}$$

Checkpoint 20 For $m = (2, 1, 1)$, write the three slab inequalities explicitly:

- One for x,
- one for y,
- one for z.

Checkpoint 21 Add the simplex constraint $x + y + z = 4$.
Explain why Webster's region for m is the intersection of:

- three rounding slabs, and
- the simplex plane.

Checkpoint 22 On the barycentric grid:

- Draw the three rounding boundary lines $x = 2.5$, $y = 1.5$, $z = 1.5$.
- Shade the region where all three Webster rounding inequalities are satisfied.

Comparison With Hamilton.

Checkpoint 23 Compare the geometry of Hamilton's regions with Webster's regions.

- Which method uses remainder comparisons?
- Which method uses fixed rounding thresholds?
- Which introduces a global scaling parameter?

Checkpoint 24 Which method do you expect to be more stable under small changes in population data? Explain your reasoning geometrically.

In this activity you explored two apportionment methods as geometric functions. Hamilton's method partitions the simplex using remainder comparisons, while Webster's method partitions quota space using rounding thresholds controlled by a global scale.